

TIANKAI YANG

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CS Department, GCS Hall

Los Angeles, CA

University of Southern California

Department of Computer Science

RESEARCH INTERESTS

1. Post-training Alignment for Trustworthy LLMs

Safety Alignment, Preference Learning

2. Robust and Reliable LLM Inference

Hallucination Mitigation, Jailbreak Detection, OOD Detection, Multimodal Robustness, Model Selection & Routing

3. Trustworthy LLM Agents and Agentic Systems

Agent Safety, Runtime Reliability, Multi-Agent Orchestration

EDUCATION

University of Southern California, Los Angeles, CA

Aug 2024 – Present

Ph.D. in Computer Science

· Affiliation: **FORTIS** Lab; Advisor: Prof. Yue Zhao

University of Southern California, Los Angeles, CA

Jan 2022 – Dec 2023

M.S. in Machine Learning and Data Science

Nankai University, Tianjin, China

Sep 2017 – Jul 2021

B.E. in Software Engineering

WORK EXPERIENCE

LinkedIn, AI/ML Engineer Intern, Generative AI

May 2026 – Aug 2026

User Simulation Agent, LLM Reasoning, Post-training

Sunnyvale, CA

- Developing the reasoning component of an LLM-based user-simulation agent to improve user-behavior prediction, via post-training with GRPO and OPSD.

SELECTED PUBLICATIONS

* denotes joint first authors. See [Google Scholar](#) for the full publication list.

- **Cat-DPO: Category-Adaptive Safety Alignment**

Tiankai Yang*, Yi Nian*, Xinyuan Li, Ruiyao Xu, Henry Peng Zou, Kaize Ding, Xiyang Hu, Yan Liu, Yue Zhao.
NeurIPS, 2026

- **HyperSkill: Self-Evolving LLM Agents via Hypergraph-Structured Skill Memory**

Ruiyao Xu, Tiankai Yang, Wei-Chieh Huang.
NeurIPS TTCL Workshop, 2026

- **SynCo: Data Synthesis Co-Training for Self-Evolving LLMs via Multi-Agent Reinforcement Learning**
Wei Yang, Li Li, Yuehan Qin, Yawei Wang, Mingxi Wang, Shixuan Li, Tiankai Yang, Jiate Li, Jesse Thomason, Xuezhe Ma, Yue Zhao.
NeurIPS TTCL Workshop, 2026
- **CoAct: Co-Active LLM Preference Learning with Human-AI Synergy**
Ruiyao Xu, Mihir Parmar, Tiankai Yang, Zhengyu Hu, Yue Zhao, Kaize Ding.
ACL, 🏆 Oral, 2026 🔄
- **DOG-DPO: Dynamic Optimization in Geometry for Safety Alignment**
Yi Nian, Tiankai Yang, Yudi Zhang, Qi Pan, Zelong Xu, Shenzhe Zhu, Qingqing Luan, Yue Huang, Xiangliang Zhang, Yue Zhao.
EMNLP Findings, 2026
- **FlexRouter: Learning Complementary Model Sets for Flexible LLM Routing**
Wang Wei, Harry Yang, Tiankai Yang, Samyadeep Basu, Hongjie Chen, Andy Zhao, Franck Dernoncourt, Ryan A. Rossi, Hoda Eldardiry.
COLM, 2026
- **AD-AGENT: A Multi-agent Framework for End-to-end Anomaly Detection**
Tiankai Yang*, Junjun Liu*, Michael Siu*, Jiahang Wang, Zhuangzhuang Qian, Chanjuan Song, Cheng Cheng, Xiyang Hu, Yue Zhao.
IJCNLP-AACL Findings, 2025 🔄
- **AD-LLM: Benchmarking Large Language Models for Anomaly Detection**
Tiankai Yang*, Yi Nian*, Shawn Li, Ruiyao Xu, Yuangang Li, Jiaqi Li, Zhuo Xiao, Xiyang Hu, Ryan A. Rossi, Kaize Ding, Xia Hu, Yue Zhao.
ACL Findings, 2025 🔄
- **DPU: Dynamic Prototype Updating for Multimodal Out-of-Distribution Detection**
Shawn Li, Huixian Gong, Hao Dong, Tiankai Yang, Zhengzhong Tu, Yue Zhao.
CVPR, 🏆 Highlight, 2025 🔄
- **PyOD 2: A Python Library for Outlier Detection with LLM-powered Model Selection**
Sihan Chen*, Zhuangzhuang Qian*, Michael Siu*, Tiankai Yang, Xingcan Hu, Jiaqi Li, Shawn Li, Yuehan Qin, Zhuo Xiao, Wanghao Ye, Yichi Zhang, Yushun Dong, Yue Zhao.
The Web Conference (Demo Track), 2025 🔄

SELECTED PREPRINTS & UNDER SUBMISSION

- **Model Routing for Large Language Models: A Unified Taxonomy and Survey**
Wang Wei, Tiankai Yang, Xiangbo Gao, Chan-Wei Hu, Reuben Luera, Runzhou Liu, Nigel Fernandez, Li Li, et al.
Under submission. SSRN preprint 🔄
- **No Attacker Needed: Unintentional Cross-User Contamination in Shared-State LLM Agents**
Tiankai Yang, Jiate Li, Yi Nian, Shen Dong, Ruiyao Xu, Ryan A. Rossi, Kaize Ding, Yue Zhao.
Under submission. arXiv preprint arXiv:2604.01350

SERVICES

Program Committee for Conferences

- NeurIPS 2026; ACL ARR 2025 (*Outstanding Reviewer*), 2026; AAI 2026, 2027; COLM 2026; ICLR 2026, 2027; ICML 2025, 2026.

Journal Reviewer

- IEEE Transactions on Computational Social Systems